Monologue or Dialogue?

Measuring Conversational Structure in the Reasoning Traces of Large Language Models

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Abstract

A reasoning model such as DeepSeek-R1 emits a long "thinking" trace before it answers. That trace can read as one mind reasoning step by step, or as several voices trading proposals and objections. We ask which description the *text* fits when the question is put to measurement. Because no reasoning trace comes labelled with its true voice structure, we build a 2×2 grid of corpora whose structure *is* known — human and AI, monologue and dialogue — and admit a detection method to the traces only once it can tell monologue from dialogue on the four known corpora. Three methods qualify, each reading a different signal: a document-level author-count detector, a sentence-level style-switch detector, and a dialogue-act detector that measures adjacency-pair completion. Applied to the traces, *two of the three* place them on the dialogue side, past the dialogue pole — the sentence-level style switching and the adjacency-pair completion — while the coarse document-level author-count detector leans monologue, and only marginally. Read together, R1 traces carry dialogue-like structure both in their sentence-to-sentence style and in their question-and-answer organisation; only the bluntest measure, whole-document authorship, fails to resolve them into distinct voices.

1 The question

Reasoning-oriented training is now standard in frontier models: given a problem, a model like DeepSeek-R1 [1] first generates an extended stretch of intermediate tokens [2] — a reasoning trace — and only then commits to an answer. These traces have a curious property. Each is produced by a single model in one pass, yet a trace often contains questions and their answers, objections and concessions, and abrupt reversals ("Wait, that's not right"). In human writing such features can mark a genuine exchange, but they can equally mark one vivid inner voice. Hence a question easy to state and less easy to settle: *does a reasoning trace have the textual structure of a monologue or of a dialogue?*

The question has a live theoretical stake. Kim et al. [3] argue that reasoning-trained models improve partly because they simulate a "society of thought" — distinct internal perspectives that propose, criticise, and reconcile — and older ideas echo it: reasoning as at root argumentative and social [4], and the mind as many interacting agents [5]. Our premise is narrower and testable: voice structure leaves marks in the writing itself. If distinct perspectives take turns producing the text, they should also differ in the small, largely involuntary habits — function-word use, sentence rhythm, dialogue-act moves — that authorship and conversation analysis use to tell writers and turns apart. We therefore treat the question operationally: a monologue should read as one author, a dialogue as several voices exchanging turns, and we ask which the traces resemble.

2 The design: a calibration grid

No corpus of reasoning traces is labelled with its true voice structure, which is what makes the question hard to settle directly. We address this with the grid in Figure 1. Four corpora of *known* structure vary producer (human vs. AI) and structure (monologue vs. dialogue): human monologue (think-aloud protocols [6]), human dialogue (group deliberation, DeliData [7]), AI monologue (one Qwen-2.5-7B instance solving a puzzle alone), and AI dialogue (four stateless Qwen-2.5-7B instances, each given a fixed role, debating). The fifth cell — native DeepSeek-R1 traces — is the target.

The grid works as a *gate*: a method may report on the target only after it correctly separates monologue from dialogue on all four known cells. A detector that cannot tell voices apart in a genuine four-way debate is in no position to rule them out in a trace; one that reports several voices in a lone think-aloud protocol is in no position to affirm them there. This is a validity check the method must earn, not an assumption we grant it.

One design choice keeps the contrast clean: we hold the task fixed wherever possible. `ai-mono`, `ai-dia`, `ai-reason` and `hum-mono` all use the same puzzle, Game of 24 [8], so a method reading a monologue as one voice cannot merely be reacting to subject matter. (`hum-dia` uses the Wason task, the closest human group-deliberation data available; its role is only to confirm that a method detects human dialogue at all, so it never anchors the placement.) Speaker-identity tokens — pseudonyms, role titles, letter labels — were stripped from both dialogue cells before analysis, so dialogue detection cannot be an artefact of speaker cues rather than interactional structure.

3 Three methods, three signals

The methods read deliberately different signals, so that agreement among them carries more weight than any one alone.

Method 1 — document-level author count [9]. Clusters a document's paragraph-sized windows on their 50-most-frequent-token profiles and reports the number of clusters as

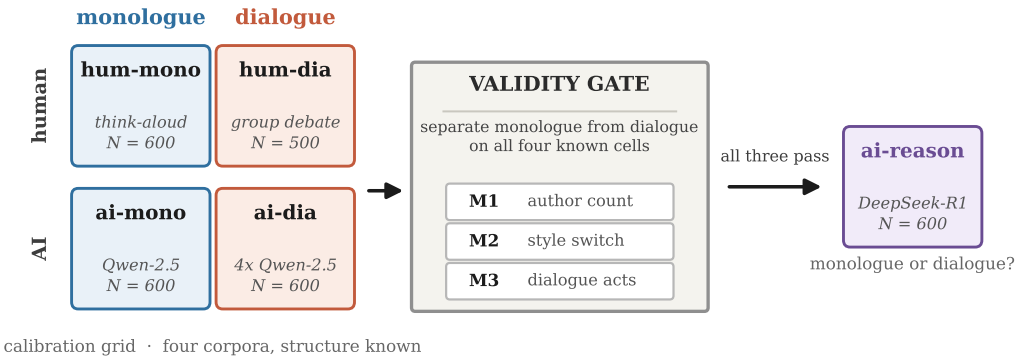


Figure 1: The design. Left to right: four corpora of known voice structure feed a *validity gate* that each detection method must pass — separating monologue from dialogue on all four known cells. All three methods qualify and are then applied to the DeepSeek-R1 traces (`ai-reason`), which they place on the monologue–dialogue axis anchored by the two same-task AI cells; the human cells validate only.

an author count, reduced to a binary single/multi call. *Gate:* balanced accuracy across the four cells must beat the 0.50 chance baseline.

Method 2 — sentence-level style switch [10]. A pre-trained sequential sentence-pair classifier emits, for each adjacent sentence pair, the probability that the two were written by different authors, using content-independent style embeddings [11]. Because raw scores depend on length, each document's score is turned into a length-adjusted *lean* — its distance to the dialogue anchor's fitted length curve minus its distance to the monologue anchor's. *Gate:* mean lean must be higher on the dialogue cells than the monologue cells, with Cliff's $\delta \geq 0.33$.

Method 3 — dialogue-act structure [12, 13]. An LLM tags each utterance with a CAMS dialogue act and measures adjacency-pair completion: when a first pair-part (a question, request or offer) occurs, is it met by a valid second pair-part (an answer, a confirmation, an acceptance) in the very next utterance — the mark of conversation a lone writer has little reason to produce. *Gate:* the completion rate must separate the dialogue cells from the monologue cells, with Cliff's $\delta \geq 0.33$.

4 Results

Method 1 passes, but softly. Balanced accuracy is 0.6088 (bootstrap 95% CI $[0.596, 0.621]$), clearing the chance baseline. But the pass leans on near-perfect single-author recall (0.975 on `hum-mono`, 0.998 on `ai-mono`) averaged against weak multi-author recall: 0.423 on `ai-dia` and a complete miss on `hum-dia` (0.000). The detector flags dialogue where it finds it but essentially never rules single authorship *out*. A monologue-leaning placement under this method should thus be read as the *absence* of a detected authorship signal, not positive evidence of one voice.

Method 2 passes cleanly. Mean lean is higher on the dialogue cells than the monologue cells (0.0125 vs. −0.0105) with Cliff's $\delta = 0.6211$, well above the 0.33 floor. The traces themselves (0.0433) sit above *both* AI anchors.

Table 1: Cross-method verdicts on the reasoning traces. This is the answer to the research question.

Method	Passed gate?	Verdict on `ai-reason`
1: author count (doc.)	pass	monologue (marginal)
2: style switch (sent.)	pass	dialogue
3: adjacency pairs	pass	dialogue

Method 3 passes. Its adjacency-pair completion rate separates dialogue from monologue in the expected direction, with Cliff's $\delta = 0.4717$, above the 0.33 floor: within each producer the dialogue cell completes pairs more often than the monologue cell (human 0.235 vs. 0.096; AI 0.025 vs. 0.003). Its two descriptive signals — the distribution of dialogue-act tags and their transitions — do not separate the AI poles, because every cell is dominated by the `inform` act; the informative signal is completion, on which the method rests.

Two of the three point to dialogue (Table 1, Figure 2). Method 2, reading sentence-to-sentence style, places the traces firmly on the *dialogue* side, past the dialogue anchor itself. Method 3, reading the completion of question-and-answer pairs, places them past the dialogue anchor as well: the trace completion rate (0.035) sits closer to the dialogue anchor `ai-dia` (0.025) than to the monologue anchor `ai-mono` (0.003). Only Method 1, the coarsest measure, leans *monologue*, and it does so marginally — its marker sits at the very edge of a wide no-decision band. The two methods that read fine-grained interactional structure agree; the disagreement is confined to the one method that reads only coarse, document-level authorship.

5 What it means

The claim this study licenses is about *textual surface structure*, not hidden minds: whether the traces exhibit the style-switching and adjacency-pair regularities of multi-party dialogue. On that question the two methods that measure

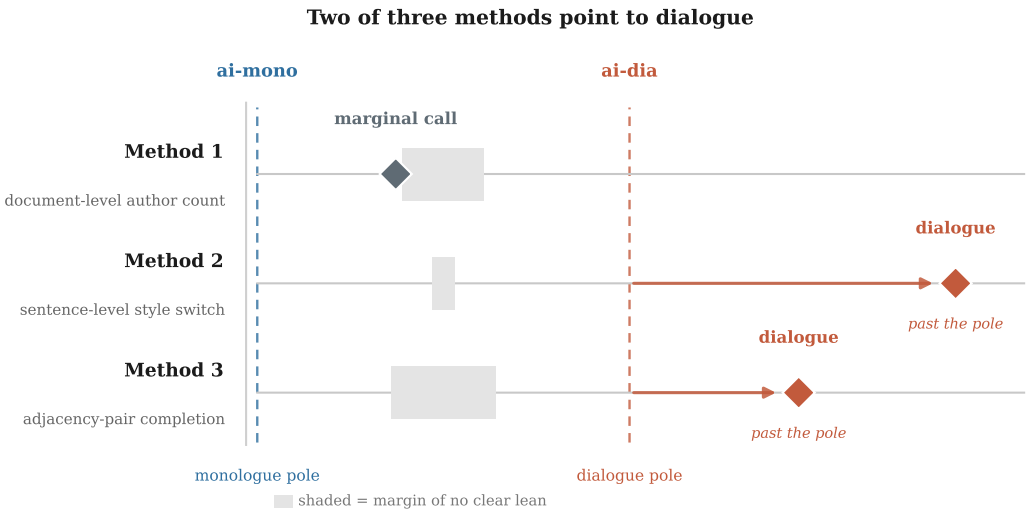


Figure 2: Where the three methods place the traces. Each method's placement of the target on the monologue–dialogue axis (poles at 0 and 1). The shaded band is that method's margin, within which no lean is called. Method 1's marker sits at the edge of a *wide* margin, just onto the monologue side — a marginal call. Methods 2 and 3 both land well past the dialogue pole. Two of the three point to dialogue.

fine-grained interactional structure agree: the traces switch style from sentence to sentence like a dialogue, and they complete question-and-answer pairs at a rate above the dialogue anchor. The lone monologue call comes from the coarsest measure, whole-document author clustering — and that method is a conservative detector whose monologue verdict means "no authorship signal found," not "positively one voice." The weight of the evidence therefore tilts toward dialogue.

This aligns the outside view with the inside one. Kim et al. [3] infer distinct internal perspectives from interpretability and trace labelling; the surface text measured here carries the matching marks — style that shifts across the trace and first pair-parts met by their second parts — which is the measurable surface counterpart of the internal multi-agent dynamics they report. A dialogue-leaning reading does not show that a model contains multiple minds, any more than a novelist's dialogue does; it shows that the trace is organised as conversation, itself a substantive, measurable fact.

Two qualifications temper the reading, and both concern how a distance to a pole is obtained rather than its direction. Method 2's monologue distance is *extrapolated*: the traces run far longer than the monologue anchor, so what the method measures directly is that they track the *dialogue* anchor closely across the shared length range, while the monologue comparison extends a length curve past the data that fit it. And Method 3's verdict rests specifically on the adjacency-pair completion signal; its coarser tag profiles cannot separate the AI poles at all. Neither qualification reverses a direction — both dialogue-ward placements survive them — but each bounds how far past the pole the placement should be read.

6 Limitations

The target is one model (DeepSeek-R1) on one puzzle family, so the verdict is about this cell, not reasoning traces at large. The target is also a *different* base model than the anchors (Qwen-2.5-7B), so any measured distance carries a shared model offset; the placement is best read as "closer to which pole," not as an absolute distance. Method 3's dialogue-act labels are produced by an automatic tagger, and Method 2's monologue comparison rests on extrapolation beyond the fitted length range. These are the reasons the validity gate matters: a method that survives it has at least shown it can tell the two structures apart on data where the answer is known.

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